Table 1: Reconstruction quality on RippleBench-Synthetic (64 sensors, 256×256 fields, mean over 5 training runs). Higher PSNR/SSIM is better; lower runtime is better.

Method	PSNR (dB) ↑	SSIM ↑	Runtime (ms) ↓
Linear interpolation	21.37	0.612	**0.8**
Radial-basis interpolation	24.05	0.701	6.4
Quillwave-Lite (ablation)	27.82	0.834	3.1
Quillwave (ours)	**29.46**	**0.887**	4.7

Table 2: PSNR (dB) on RippleBench-Synthetic as the number of sensors changes (256×256 fields).

Method	16 sensors	32 sensors	64 sensors	128 sensors
Linear interpolation	16.92	19.10	21.37	23.88
Radial-basis interpolation	18.44	21.26	24.05	26.71
Quillwave-Lite (ablation)	21.03	24.67	27.82	30.15
Quillwave (ours)	**22.58**	**26.31**	**29.46**	**31.94**

Quillwave: Sparse-Sensor Reconstruction of Synthetic Ripple Fields

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Project summary. Research in preparation; not peer reviewed.

Abstract

Quillwave reconstructs a full 256×256 ripple field from 64 point sensors. Ripple fields are procedurally generated sums of interfering circular waves: cheap to synthesise, but hard to interpolate, because the wave fronts between sensors are lost by smooth interpolation. Quillwave encodes the sensor readings into a 128-dimensional latent field code and decodes it with a coordinate-based field decoder. On RippleBench-Synthetic, a benchmark of 500 synthetic test fields generated for this work, Quillwave reaches 29.46 dB PSNR and 0.887 SSIM, 5.41 dB above radial-basis interpolation (the best baseline), which is a 71.2 % lower mean squared error. It is also 1.36× faster than radial-basis interpolation at inference. All results are on synthetic fields only.

Results

Table 1 gives the main result. Quillwave reaches 29.46 dB PSNR and 0.887 SSIM, 5.41 dB above radial-basis interpolation, the best baseline; in terms of mean squared error this is a 71.2 % reduction. At 4.7 ms per field it is 1.36× faster than radial-basis interpolation (6.4 ms), though slower than linear interpolation (0.8 ms). The ablation Quillwave-Lite reaches 27.82 dB, so the latent field code accounts for part of the gain.

Table 2 varies the number of sensors. Quillwave is the best method at every sensor count, from 22.58 dB with 16 sensors to 31.94 dB with 128 sensors. Figure 1 summarises these trends.

Limitations

- **Baselines are not compute-matched.** Quillwave is trained for 3.5 GPU-hours, while the interpolation baselines need no training at all.

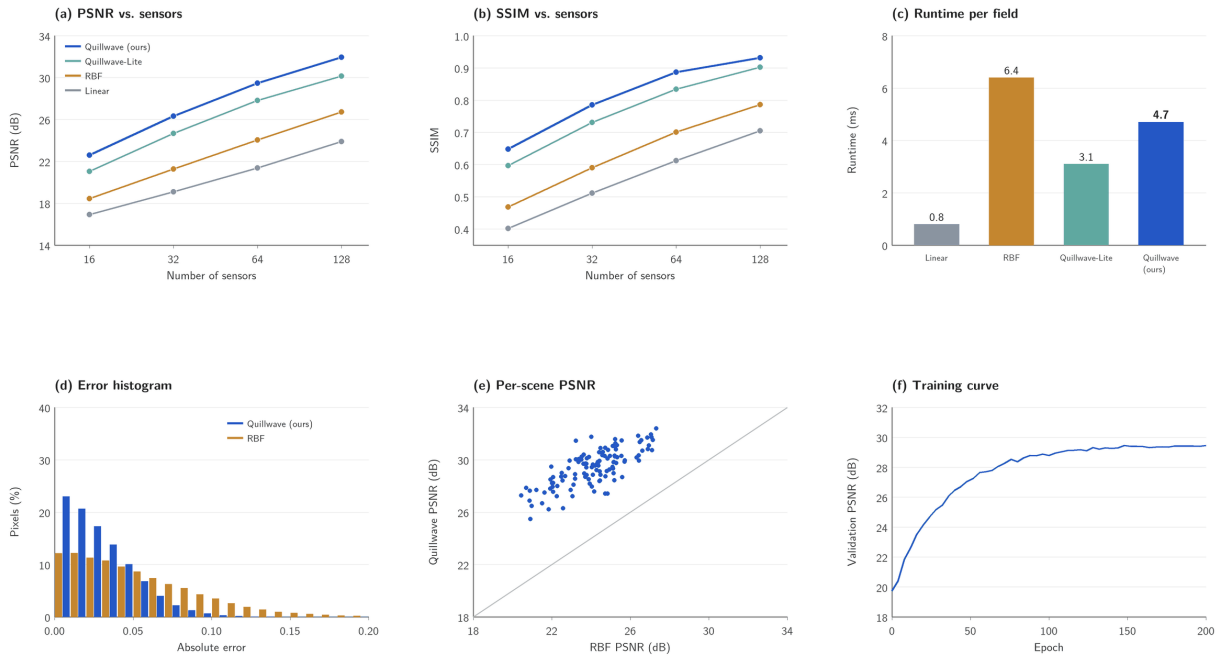


Figure 1: Results on RippleBench-Synthetic. (a) PSNR and (b) SSIM against the number of sensors; (c) runtime; (d) absolute error histogram; (e) per-scene PSNR, Quillwave against radial-basis interpolation; (f) validation PSNR during training.

- **Synthetic fields only.** Every result is on RippleBench-Synthetic; no real sensor data was used, and behaviour on real measurements is unknown.
- **Confidence is not calibration.** The per-pixel confidence map that Quillwave outputs has not been checked against observed error rates, so it should not be read as a calibrated probability.